AIDA Publication Pipeline

Operator manual

Step by step instructions for running, checking and maintaining the scripts that build the AIDA master bibliography from the harvests and from the authors' own confirmed lists, and that turn it into publication pages, author and affiliation pages, Excel workbooks, figures and review packets.

Item	Value
Project	AIDA cloud chamber bibliometrics, IMK-AAF, KIT
Prepared by	Loghman Fathollahi (loghman.fathollahi@kit.edu)
Audience	Deputy operator of the pipeline
Version	2.0, 26 October 2026 (replaces 1.0 of 29 September 2026)
Scripts covered	config.py; 1_a to 1_d; 2_a to 2_c for Ottmar Möhler and Robert Wagner; 3_a_aida_master_bib, 3_a_aida_bib_html; 3_b_aida_bib_authors, 3_b_aida_bib_excel, 3_b_aida_bib_figures; 3_c; 3_d
Platforms	Linux, macOS and Windows, no path editing needed

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What changed since version 1.0

- **Runs on every operating system.** config.py finds the project root from its own location (the folder above scripts/). Set the variable AIDA_ROOT only to keep the data elsewhere. Every script puts its own folder on the import path, so it finds config.py from any working directory.
- **New phase 2: the authors' own lists.** Ottmar Möhler sent his full list with the AIDA papers marked in green (13 October). Robert Wagner sent notes on the list we sent him (14 October). Six new scripts turn both into checked bib files with a DOI for every paper.
- **The authors' lists now decide first.** The master script (renamed 3_a_aida_master_bib.py) applies the confirmed lists before the filters: confirmed papers are always kept and marked aida_confirmed_by, removed papers are always removed, and a complete list rejects unlisted papers of that author.
- **New output layout.** AIDA_master/ now has three folders: master/ (the final file), authors/ (one master per researcher) and supplementary/ (all reports).
- **New outputs.** One publication page per researcher with an index page, Excel workbooks (3_b_aida_bib_excel.py) and PDF or PNG figures and maps (3_b_aida_bib_figures.py).
- **Fixed from the 1.0 list.** The /home/lf paths are gone, the duplicate require_kit_affiliation setting was renamed, and the missing config names (BASE_DIR, CONTACT_EMAIL, KIT_ROR, the review folders, the noAIDA and yesAIDA files) were added. fill_corrections.py is no longer part of the set.
- **Still open.** See section 10: 1_b sends no OpenAlex key, yesAIDA.txt is still not read by the master script, and filter 4 is still redundant next to filter 1.

1. What the pipeline does

The goal is one complete, defensible list of every publication that used the AIDA cloud chamber. The list is built from bibliographic sources, corrected by the authors themselves, and then published as pages, tables and figures.

1.1 The three phases

Phase	Scripts	Result
1 Harvest	1_b (OpenAlex), manual Web of Science export, 1_d (IMK-AAF page). Checks: 1_a (WoS), 1_c (OpenAlex)	Raw .bib files under data/
2 Authors' own lists	2_a, 2_b, 2_c for Ottmar Möhler, then the same three for Robert Wagner	One confirmed bib file per author in data/author_self_correction/bib/
3 Master and outputs	3_a_aida_master_bib, 3_a_aida_bib_html, 3_b_aida_bib_authors, 3_b_aida_bib_excel, 3_b_aida_bib_figures, 3_c, 3_d	AIDA_Master_bib.bib, HTML pages, CSV rankings, Excel workbooks, figures, review packets

1.2 Data flow



Important. 1_a, 1_c and 1_d write to master_bibs/, which the master script does NOT read. They are quality reports and a reference list. The master script reads the raw harvest folders and the authors' bib files directly.

2. Before you start

2.1 Folder and Python

Keep all scripts and config.py together in one folder called scripts/ inside the project root. The root is found automatically. Python 3.9 or newer is needed.

	Linux / macOS	Windows (PowerShell)
Go to the scripts	cd ~/AIDA/scripts	cd D:\AIDA\scripts
Activate the environment	source ~/AIDA/aida/bin/activate	..\aida\Scripts\Activate.ps1
Check Python	python --version	python --version

Install the packages once. The standard library covers everything else.

```
python -m pip install requests beautifulsoup4 lxml python-docx openpyxl matplotlib
```

Package	Needed by
requests, beautifulsoup4, lxml	1_d (lxml optional, the built in parser is used without it)
python-docx	all 2_a and 2_b scripts (reading and writing Word files)
openpyxl	3_b_aida_bib_excel.py
matplotlib	3_b_aida_bib_figures.py

2.2 Environment variables

Variable	Needed by	What for
AIDA_ROOT	optional, all	Only to keep the data outside the folder above scripts/
OPENALEX_API_KEY	3_b_aida_bib_authors	OpenAlex asks for a free key since February 2026 (openalex.org/settings/api). Without it run 3_b with --no-openalex.
OPENALEX_MAILTO	3_b_aida_bib_authors	Optional, falls back to openalex_mailto in config.py
AIDA_DB_USER, AIDA_DB_PASSWORD	archive scripts only	Not needed for this manual

	How to set it
Linux / macOS	export OPENALEX_API_KEY=your_key (put it in ~/.bashrc to keep it)
Windows cmd	set OPENALEX_API_KEY=your_key
PowerShell	$env:OPENALEX_API_KEY = "your_key"

2.3 Check the configuration

Run this first on a new machine and after every change to config.py. It prints the root, every project path with ok or MISSING, the filter rules and whether the keys are set.

```
python config.py
```

2.4 Rules

- All paths and rules live in config.py. Never type a path into a script.

- Almost every script has --dry-run. Use it the first time and whenever you are unsure.
- On Windows, close a CSV, Word or Excel file before a run that rewrites it. The scripts stop with a clear message when a file is still open.
- CSV files are written as UTF-8 with BOM, so Excel shows umlauts correctly. A CSV saved back by Excel (cp1252, or ; as separator) is read correctly too.

3. Folder layout

<root> is the project root. Folders are created when a script needs them.

Folder or file	Written by	Content
scripts/	Loghman	All scripts and config.py
data/AIDA Scientists.txt	Loghman	One researcher per line, First Last. Read by 1_b, the master, Excel and figures
data/webofscience/keywords/, researchers/	Loghman	Web of Science exports
data/openalex/researchers/	1_b	One .bib per researcher; fetch_log.csv one level up
data/imkaaf/	1_d	Cached KITopen records
data/AIDA_campaigns/	archive scripts	Campaign CSVs for the campaign filter
data/noAIDA.txt	Loghman, 3_d	Papers never to appear in the master
data/author_self_correction/raw/	Loghman	The files exactly as the authors sent them. Never edit them
data/author_self_correction/	2_a, 2_b	Extracted Word lists, DOI tables, correction tables
data/author_self_correction/bib/	2_c	Ottmar_Moehler.bib, Robert_Wagner.bib
data/author_self_correction/cache/	2_b, 2_c	Crossref and BibTeX answers, so reruns need no internet
AIDA_lists/publications/AIDA_Publication_List.html	Loghman (frozen copy)	The page that was sent to Robert. Read by his 2_a and 2_c. Never overwrite it
master_bibs/, processed_data/	1_a, 1_c, 1_d	Source reports
AIDA_master/master/	3_a_aida_master_bib	AIDA_Master_bib.bib, the master
AIDA_master/authors/	3_a_aida_master_bib	One master per researcher, <First_Last>.bib
AIDA_master/supplementary/	3_a_aida_master_bib	Stage files and all reports
output/publications/	3_a_aida_bib_html	Main page, AIDA_Authors_Index.html, authors/<First_Last>.html
output/authors/	3_b_aida_bib_authors, Loghman	Author, affiliation and map pages, CSVs, cache/, hand edited alias files
output/excel/	3_b_aida_bib_excel	AIDA_Master.xlsx, authors/<First_Last>.xlsx
output/figures/	3_b_aida_bib_figures	AIDA_Master/ and authors/<First_Last>/, with Notes.txt
info/aut/review/	3_c	authors_registry.csv, _index.csv, one packet per author
info/aut/corrections/	Loghman, 3_d	Returned packets; _merged/ holds the proposals

4. Scripts at a glance

Script	In one line	Net	Run when
config.py	Central settings; python config.py checks them	no	after any config change
1_a_aida_wos_merge.py	Four stage quality report of the WoS exports	no	after new exports
1_b_aida_openalex.py	OpenAlex harvest, one .bib per researcher	yes	new names, yearly refresh
1_c_aida_openalex_merge.py	Same report for the OpenAlex harvest	no	after 1_b
1_d_aida_imkaaf_bib.py	BibTeX of every paper on the IMK-AAF page	yes	reference check
2_a_aida_ottmar_list.py	Keeps the green (AIDA) entries of Ottmar's Word list	no	once, or when he sends a new list
2_b_aida_ottmar_check.py	Finds and checks a DOI for each of his entries	Crossref	after 2_a, after hand fixes
2_c_aida_ottmar_bib.py	Builds Ottmar_Moehler.bib from the DOI table	doi.org	after 2_b
2_a_aida_robert_list.py	Writes Robert's entries of the sent page as a Word list	no	once
2_b_aida_robert_check.py	Applies his notes, completes the DOIs	Crossref	after 2_a and Ottmar's 2_b
2_c_aida_robert_bib.py	Builds Robert_Wagner.bib	doi.org	after 2_b and Ottmar's 2_c
3_a_aida_master_bib.py	Builds the master and one master per researcher	no	after any phase 1 or 2 change
3_a_aida_bib_html.py	Publication page, one page per researcher, index	no	after the master
3_b_aida_bib_authors.py	Author, affiliation and map pages, CSV rankings, cache	OpenAlex, ROR	after the master
3_b_aida_bib_excel.py	Excel workbooks per researcher and for all of AIDA	no	after 3_b_aida_bib_authors
3_b_aida_bib_figures.py	PDF or PNG plots and maps	once, outlines	after 3_b_aida_bib_authors
3_c_aida_author_packets.py	One review folder per author	no	before a review round
3_d_aida_merge_corrections.py	Reads returned packets, writes proposals	no	when answers arrive

Order inside a letter. Two scripts share 3_a and three share 3_b. Run 3_a_aida_master_bib.py before 3_a_aida_bib_html.py, and 3_b_aida_bib_authors.py before the Excel and figures scripts. Ottmar's 2_b and 2_c must run before Robert's, because Robert's scripts read Ottmar's DOI table and cache.

5. Phase 1: harvesting

Step 1. Export from Web of Science (manual)

Item	Detail
Script	none, done in the browser
Purpose	WoS records carry author addresses and the corresponding author marker.
Reads	webofscience.com (KIT licence, KIT network or VPN)
Writes	data/webofscience/keywords/*.bib and data/webofscience/researchers/*.bib
Network	yes, browser
Needed	When new papers are expected

1. Search Web of Science Core Collection, for example "AIDA" AND aerosol (topic) or Mohler O* (author).
2. Export, BibTeX, record content Full Record. At most 1000 records per file, so split large results.
3. Name the file after the search and save it in keywords/ or researchers/. Do not edit it.

Step 2. Harvest OpenAlex (1_b)

Item	Detail
Script	1_b_aida_openalex.py
Purpose	Resolve every researcher to KIT affiliated OpenAlex author ids and download their works.
Reads	data/AIDA Scientists.txt
Writes	data/openalex/researchers/<name>.bib, data/openalex/fetch_log.csv
Network	yes, api.openalex.org
Needed	New names, or a yearly refresh

```
python 1_b_aida_openalex.py --dry-run       # counts only
python 1_b_aida_openalex.py                 # existing files are skipped
python 1_b_aida_openalex.py --overwrite     # refresh every file
```

Status in fetch_log.csv	Meaning	What to do
kit_match	found with a KIT affiliation	nothing
skipped	file already there	--overwrite to refresh
not_found, name_mismatch	no fitting author	check the spelling
no_kit_match	found, never at KIT in OpenAlex	rerun with --loose-authors and check by hand
http_xxx, network_error	request failed	rerun later; see section 10 on the API key

Last run. Schöck, Nink and Schäfers gave no_kit_match; for them only Web of Science counts.

Step 3. Source reports and IMK-AAF list (1_a, 1_c, 1_d; optional)

Item	Detail
Script	1_a_aida_wos_merge.py, 1_c_aida_openalex_merge.py, 1_d_aida_imkaaf_bib.py

Item	Detail
Purpose	Report per file and per year what deduplication, the keyword test and the KIT test remove; download the IMK-AAF list as an independent reference.
Reads	the harvest folders; imkaaf.kit.edu/60.php
Writes	master_bibs/*_master.bib, processed_data/wos/ and processed_data/openalex/
Network	1_d only
Needed	After new exports, to judge a search

```
python 1_a_aida_wos_merge.py
python 1_c_aida_openalex_merge.py
python 1_d_aida_imkaaf_bib.py --dry-run
python 1_d_aida_imkaaf_bib.py
```

Read wos_merge_summary.txt and openalex_merge_summary.txt. Each row reads as a funnel: records minus dup minus no keyword minus no KIT equals final. A file with almost no final records is a search to change. The keyword and KIT rules are in the WEB OF SCIENCE FILTER RULES block of config.py.

6. Phase 2: the authors' own lists

An author knows best which papers are AIDA work. Phase 2 turns what an author sent into a bib file with a checked DOI for every paper. The master script then uses these files before its filters. Each author has three steps: list, check, bib.

	Ottmar Möhler	Robert Wagner
What he sent	His full publication list (Word), AIDA papers highlighted green, 13 Oct 2026, 142 papers	Free text notes on the AIDA page we sent him, 14 Oct 2026
Raw file in raw/	Ottmar_Möhler_publications.docx (ö, oe or _ all work)	Robert_Wagner_publications.docx
Also needed	nothing	AIDA_lists/publications/AIDA_Publication_List.html, the exact page he saw
Type of list	complete: unlisted articles of his are rejected	review: his removals and additions count, new papers follow the filters

Step 4. Ottmar: extract, check, build

Item	Detail
Script	2_a_aida_ottmar_list.py, 2_b_aida_ottmar_check.py, 2_c_aida_ottmar_bib.py
Purpose	Keep his green entries, give each a DOI, build Ottmar_Moehler.bib
Reads	raw/Ottmar*publications*.docx, the harvest folders, Crossref, doi.org
Writes	Ottmar_Moehler_AIDA_publications.docx, Ottmar_Moehler_AIDA_dois.csv, bib/Ottmar_Moehler.bib, cache/
Network	Crossref (2_b), doi.org (2_c); both cached
Needed	Once, then after every hand fix in the DOI table

```
python 2_a_aida_ottmar_list.py --dry-run   # entries per colour, must say 142
python 2_a_aida_ottmar_list.py
python 2_b_aida_ottmar_check.py
python 2_c_aida_ottmar_bib.py
```

What 2_a reports

Entries per highlight colour and the AIDA count. If the count is not 142 (ottmar_expected_aida in config.py) a WARNING appears: check the highlights in his file. A paragraph with two colours is listed as a note.

How 2_b finds a DOI

Order	Method	Accepted when
1	doi_in_list	the DOI is written in his citation
2	own_harvest	a harvested record with the same first author, year and title
3	crossref	Crossref result whose first author, year and title agree
4	not_found	nothing reliable; the DOI stays empty

A title must cover at least 90 percent (doi_title_accept) for status found, 80 percent (doi_title_check) gives status check. Discussion papers (ACPD, AMTD, EGUsphere and so on) are used only when no final article exists, always as check.

Fixing by hand, the important loop

1. Open Ottmar_Moehler_AIDA_dois.csv in Excel. Filter the status column for check and not_found.
2. For each such row look up the right DOI, write it into doi and write manual into method. To leave a row without DOI on purpose, write manual and keep doi empty.
3. Save, close Excel, run 2_b again: manual rows are never changed. Then run 2_c.

What 2_c reports

Records by source: doi_org (best), crossref_api, own_harvest, citation. Every citation entry is unverified and printed with its number: give those a DOI in the table if at all possible. Two of his numbers with one DOI become one entry with both numbers.

Options: --no-crossref (2_b) and --offline (2_c) run without internet from the cache; --refresh asks again.

Step 5. Robert: list, apply notes, build

Item	Detail
Script	2_a_aida_robert_list.py, 2_b_aida_robert_check.py, 2_c_aida_robert_bib.py
Purpose	Rebuild the list he saw, apply his notes, give each paper a DOI, build Robert_Wagner.bib
Reads	the frozen page in AIDA_lists/, raw/Robert*Wagner*.docx, Ottmar's DOI table and cache
Writes	Robert_Wagner_AIDA_publications.docx, Robert_Wagner_AIDA_dois.csv, Robert_Wagner_corrections.csv, bib/Robert_Wagner.bib
Network	Crossref and doi.org, cached
Needed	Once, then after hand fixes

```
python 2_a_aida_robert_list.py      # 53 entries expected
python 2_b_aida_robert_check.py
python 2_c_aida_robert_bib.py
```

How his notes are read

Kind of note	What happens
remove	the one entry it points at is marked removed, with a reason (supplement, dataset, discussion page, earlier version). If none or several entries match, nothing is removed and the note says check
add (Missing paper)	added as a new number, unless it is already in his list. The DOI is taken from Ottmar's table when the paper is there
question (proceedings)	decided by robert_keep_proceedings in config.py (True: kept by decision)
APC chamber papers	decided by robert_keep_apc_papers in config.py (True: added)
role	not a list change; the candidate papers are named for the author pages
not understood	action check, read his text by hand

Check Robert_Wagner_corrections.csv: every note has an applied column. Every line starting with check needs a look. The same manual DOI loop as for Ottmar applies to Robert_Wagner_AIDA_dois.csv.

> **Never regenerate the frozen page.** His notes say things like "the row above". They only make sense on the exact page he saw, so AIDA_lists/publications/AIDA_Publication_List.html must stay as it was sent. 3_a_aida_bib_html.py writes to output/publications/ and never touches it.

Adding a third author later

The 2_x scripts are written for these two authors. For anyone else the simplest route is the review packet of section 8. If an author sends a full list, make a bib file for it (copy and adapt the Ottmar scripts) and add one entry to author_confirmed_lists in config.py:

```
{"name": "First Last", "surname": "Last", "initial": "F",
 "bib": AUTHOR_SELF_CORRECTION_BIB_DIR / "First_Last.bib",
 "removed": None, "scope": "articles", "complete": True},
```

Setting	Meaning
scope "articles"	the list judges only his articles; his other entries follow the filters
scope "all"	the list judges every entry with him
complete True	he marked his whole record: an unlisted paper of his is rejected
complete False	he reviewed what we sent: new papers still follow the filters
removed	a table with rows marked removed (Robert's DOI table), or None

7. Phase 3: master and outputs

Step 6. Build the master (3_a_aida_master_bib)

Item	Detail
Script	3_a_aida_master_bib.py
Purpose	Merge, deduplicate, apply noAIDA.txt and the authors' lists, filter, and write one master per researcher
Reads	data/openalex, data/webofscience, noAIDA.txt, campaign CSVs, the authors' bib files, AIDA Scientists.txt
Writes	AIDA_master/master/, authors/, supplementary/
Network	no
Needed	After any change in phase 1 or 2, noAIDA.txt or the filter

Order of decisions

1. Stage 1, merge every .bib under the harvest folders.
2. Stage 2, join copies with the same DOI, or the same title and year without DOI; fill missing fields from the other copies; new keys year_firstauthor_journal.
3. noAIDA.txt removes a paper in every stage, with all its copies.
4. Authors' lists: confirmed papers are kept and marked aida_confirmed_by; confirmed papers no harvest found are added; removed papers are removed; for a complete list, his unlisted papers are removed. When one author confirms and another removed, the paper is kept and reported as a conflict.
5. Stage 3, the remaining papers go through the selected filters.

Filter	Keeps an entry when title, abstract or keywords contain	KIT affiliation
1	an AIDA term	no
2	a campaign name	no
3	AIDA term and campaign name	no
4	an AIDA term	yes
5	a campaign name	yes
6	AIDA term and campaign name	yes

```
python 3_a_aida_master_bib.py --dry-run --verbose
python 3_a_aida_master_bib.py
python 3_a_aida_master_bib.py --stage filter      # after editing noAIDA.txt or an author
list
python 3_a_aida_master_bib.py --filter 4 5        # override master_bib_filter
```

What to check

File in supplementary/	Look for
AIDA_Master_bib_combined_summary.txt	entries in and kept; a jump of more than a few percent needs an explanation; noAIDA items without a match
AIDA_Master_bib_author_decisions.csv	every decision of the authors' lists; filter for conflicts and for confirmed papers not found anywhere
AIDA_Master_bib_authors_summary.txt	papers per researcher, and who has none (name not understood, no paper)

File in supplementary/	Look for
AIDA_Master_bib_filter_log.csv	sort by reason; removed entries that look like AIDA work point to missing terms
AIDA_Master_bib_removed.bib	removed entries for a manual look

noAIDA.txt holds one item per line or block: a DOI alone is enough; a citation without DOI is matched by its title. Lines starting with # are comments.

Step 7. Publication pages (3_a_aida_bib_html)

Item	Detail
Script	3_a_aida_bib_html.py
Purpose	Searchable publication page with year histogram, type and confirmation filters; one page per researcher; an index
Reads	AIDA_master/master/AIDA_Master_bib.bib, AIDA_master/authors/*.bib
Writes	output/publications/AIDA_Publication_List.html, AIDA_Authors_Index.html, authors/<First_Last>.html
Network	no
Needed	After every master run

```
python 3_a_aida_bib_html.py --dry-run --verbose
python 3_a_aida_bib_html.py
python 3_a_aida_bib_html.py --no-authors     # main page only
```

Confirmed papers show "confirmed by" under the title, and a filter shows only confirmed or only unconfirmed papers. The verbose report lists entries without year, DOI, title or author, and keys with leftover LaTeX: fix those in the source data.

Step 8. Authors, affiliations and map (3_b_aida_bib_authors)

Item	Detail
Script	3_b_aida_bib_authors.py
Purpose	Authors in order, roles, affiliations and countries for every paper; three linked pages and seven CSV rankings; the cache used by Excel, figures and packets
Reads	the master, output/authors/cache/, the hand edited files below
Writes	output/authors/*.html, AIDA_*.csv, cache/
Network	OpenAlex (key), ROR, map assets once
Needed	After every master run and every 3_d --apply

```
python 3_b_aida_bib_authors.py                 # only new papers are requested
python 3_b_aida_bib_authors.py --no-openalex   # no key: bib file and cache only
python 3_b_aida_bib_authors.py --refresh-cache  # ask again for earlier misses
```

Hand edited file in output/authors/	Columns	Use
author_aliases.csv	variant,canonical	merge or split authors: "Mohler, O.","Möhler, Ottmar"
affiliation_aliases.csv	pattern,canonical	regex on the folded affiliation text
corresponding_overrides.csv	doi_or_key,author,corresponding,note	role corrections; written by 3_d --apply

Hand edited file in output/authors/	Columns	Use
affiliation_coords_manual.csv	name,lat,lon,country,city	fixes map positions

Check the terminal report (papers matched in OpenAlex, share with a known corresponding author and affiliation), the top authors (two lines for one person means a missing alias) and the map (points in the sea mean a bad position).

Step 9. Excel workbooks and figures (3_b_aida_bib_excel, 3_b_aida_bib_figures)

Item	Detail
Script	3_b_aida_bib_excel.py, 3_b_aida_bib_figures.py
Purpose	Tables and figures per researcher of AIDA Scientists.txt and for all of AIDA; the figures use the same numbers as the sheets
Reads	the master and the 3_b cache; no internet except the country outlines once
Writes	output/excel/ and output/figures/
Network	figures: one download of about 750 kB if the outlines are missing
Needed	After 3_b_aida_bib_authors

```
python 3_b_aida_bib_excel.py
python 3_b_aida_bib_excel.py --only "Robert Wagner"     # one researcher, no master workbook
python 3_b_aida_bib_figures.py                          # PDF
python 3_b_aida_bib_figures.py --format png --dpi 300
```

Workbook sheet	Content
Summary	every count, linked to its sheet
Papers	year, title, journal, DOI, his position and role, affiliation, confirmed by
Affiliations, Co-authors, Institutions, Countries, Journals	rankings
Per year	papers per year by role
Data coverage	papers missing affiliation or role data, so a gap is not read as a fact
Researchers, Authors (master workbook only)	all researchers side by side; every author

Figures per researcher: roles, per year, journals, co-authors, institutions, countries, and four maps. Each folder has Notes.txt saying how many institutions every map shows and which countries are not on the map. Researchers without a paper in the master get no file; the screen says why. Both scripts empty their output folders first, so close the files before a run.

Step 10. Review packets (3_c)

Item	Detail
Script	3_c_aida_author_packets.py
Purpose	One private folder per author: profile page, corrections.csv, missing_papers.txt, README.txt
Reads	the master, the 3_b cache and alias files
Writes	info/aut/review/

Item	Detail
Network	no
Needed	Before a review round

```
python 3_c_aida_author_packets.py --dry-run --verbose
python 3_c_aida_author_packets.py --kit-only --min-papers 3 --zip
python 3_c_aida_author_packets.py --only mohler_o --zip
```

- Ids in authors_registry.csv (aut0001 ...) are fixed forever. Never edit or reuse them; the email column may be filled in.
- A rebuild replaces the packet folder. Returned files belong in info/aut/corrections/, never in review/.
- Track each packet in _index.csv: sent_on, replied_on, status. These columns survive rebuilds.
- Change contact_line at the top of 3_c if replies should go to you.

Step 11. Merge the answers (3_d)

Item	Detail
Script	3_d_aida_merge_corrections.py
Purpose	Apply what only the author can answer; count shared questions as votes
Reads	info/aut/corrections/, the registry, the master, noAIDA.txt, yesAIDA.txt
Writes	info/aut/corrections/_merged/; with --apply also corresponding_overrides.csv, noAIDA.txt, yesAIDA.txt
Network	no
Needed	When answers arrive

1. Run without --apply: python 3_d_aida_merge_corrections.py --verbose
2. Read _merged/summary.txt. Fix every line under Problems in the returned file and run again.
3. Decide conflicts.csv with Ottmar; put a removal into noAIDA.txt by hand.
4. Work through manual_review.csv: affiliations go to affiliation_aliases.csv, identity problems to author_aliases.csv, author order needs the bib entry checked.
5. Check both proposal files, then run with --apply.
6. Rebuild: 3_a_aida_master_bib --stage filter, 3_a_aida_bib_html, 3_b_aida_bib_authors, then Excel and figures.

8. Routine checklists

8.1 Full rebuild

```
python config.py
python 1_b_aida_openalex.py
python 1_a_aida_wos_merge.py            # optional
python 1_c_aida_openalex_merge.py       # optional
python 2_b_aida_ottmar_check.py && python 2_c_aida_ottmar_bib.py
python 2_b_aida_robert_check.py && python 2_c_aida_robert_bib.py
python 3_a_aida_master_bib.py
python 3_a_aida_bib_html.py
python 3_b_aida_bib_authors.py
python 3_b_aida_bib_excel.py
python 3_b_aida_bib_figures.py
```

On Windows cmd the && works the same; in older PowerShell run the lines one by one.

8.2 After editing noAIDA.txt or a DOI table

```
python 2_c_aida_ottmar_bib.py            # only if Ottmar's table changed
python 2_c_aida_robert_bib.py            # only if Robert's table changed
python 3_a_aida_master_bib.py --stage filter
python 3_a_aida_bib_html.py
python 3_b_aida_bib_authors.py
python 3_b_aida_bib_excel.py && python 3_b_aida_bib_figures.py
```

8.3 Review round

```
python 3_c_aida_author_packets.py --kit-only --zip
#  send, track in _index.csv, save answers in info/aut/corrections/
python 3_d_aida_merge_corrections.py --verbose
python 3_d_aida_merge_corrections.py --apply
#  then 8.2 from the master step on
```

8.4 Backups

Before any --apply, --overwrite or --refresh, copy the hand maintained files. They cannot be rebuilt:

- data/noAIDA.txt, data/yesAIDA.txt, data/AIDA Scientists.txt
- data/author_self_correction/ (raw/, the DOI tables with your manual rows, cache/)
- AIDA_lists/publications/AIDA_Publication_List.html (the page Robert saw)
- output/authors/*.csv that you edited, info/aut/review/authors_registry.csv and _index.csv, info/aut/corrections/
- scripts/config.py

9. Troubleshooting

Symptom	Likely cause	Fix
python config.py shows many MISSING paths	Scripts not in a folder called scripts/, or wrong AIDA_ROOT	Move them, or set AIDA_ROOT to the project root
python-docx / openpyxl / matplotlib is missing	Package not installed	python -m pip install <name>
Cannot write ..., it is probably open	File open in Excel or Word (Windows)	Close it and run again
2_a Ottmar: WARNING count differs from 142	A highlight is missing or has another colour	Check his file; add the colour to ottmar_aida_highlight_colors if needed
2_a Ottmar: More than one possible list	Two matching files in raw/	Keep one, or use --input
2_b: many check and not_found rows	Short or unusual citations	The manual loop of step 4
2_c: unverified, built from the citation	No DOI for that row	Add a DOI by hand in the table, rerun 2_b and 2_c
2_b Robert: note not applied: check	The note matches none or several entries	Read his text; mark the row by hand in his DOI table
Robert scripts: Ottmar's DOI list not found	Ottmar's 2_b not run yet	Run Ottmar's steps first
Master: author list conflict	One author confirms, another removed	Decide with Ottmar; noAIDA.txt or a list correction
Excel or figures: WARNING no OpenAlex cache	3_b_aida_bib_authors not run	Run it first
Figures: institution maps skipped	No coordinates in the cache	Run 3_b_aida_bib_authors with internet once
Researcher gets no workbook or figures	No paper in the master, or name not First Last	Check AIDA_Master_bib_authors_summary.txt
1_b: http_401 or http_403	OpenAlex now wants a key	See section 10, item 1
Same person twice in a ranking	Missing alias	author_aliases.csv, rerun 3_b
Umlauts wrong in a CSV	Opened as Latin-1	Data, From Text/CSV, UTF-8

10. Known issues and recommended improvements

Found while reading the October scripts. Items 1 to 3 can give an incomplete result without any error.

#	Priority	Issue	Recommended change
1	High	1_b still sends no OpenAlex API key (only mailto), while 3_b does. Harvests without a key may be refused or throttled.	Read OPENALEX_API_KEY in 1_b as 3_b does and add api_key to every request.
2	High	yesAIDA.txt is written by 3_d but still not read by the master script. Missing papers named in packets never reach the master. (Papers in the authors' confirmed lists are added, so this only affects packet answers.)	Read YES_AIDA_FILE in 3_a_aida_master_bib.py and treat it like a confirmed list without an author: keep, and add when no harvest has it.
3	High	master_bib_filter = [1, 4, 5]: filter 4 is inside filter 1, and filter 1 lets in unrelated AIDA acronyms (medicine, reactor, economics) because it has no KIT check.	Decide [4, 5] or [1, 5] and write the reason in config.py; add the known false hits to noAIDA.txt.
4	Medium	Two scripts share 3_a and three share 3_b, so the run order is not visible from the names.	Renumber, for example 3_a master, 3_b html, 3_c authors, 3_d excel, 3_e figures, 4_a packets, 4_b merge; update the author scripts' loading paths.
5	Medium	Old names in messages: 3_d says "Now rebuild: aida_master_bib.py"; 1_a and 1_c headers say aida_wos_merge.py and aida_openalex_merge.py; the master log says aida_master_bib.py.	Use the real file names, best from Path(__file__).name.
6	Medium	Dates and source texts are fixed in code: aida_source_text in both 2_c, decision_date in Robert's 2_b, the Ottmar line in the 3_d summary, contact_line in 3_c.	Move them to config.py next to the other author settings.
7	Medium	3_a_aida_bib_html, 3_b_aida_bib_authors, 3_c and 3_d fall back to AIDA_master/AIDA_Master_bib.bib when config.py is missing; the master now lives in AIDA_master/master/.	Change the fallback to MASTER_DIR / "master" as in 3_b_aida_bib_excel.
8	Low	The IMK-AAF list (1_d) is not compared with the master.	Add a short report to the master listing IMK-AAF DOIs missing from it.
9	Low	Output name AIDA_Master_bib_first_satge.bib has a typo.	Rename to ..._first_stage.bib in config.py.
10	Low	No single command runs the chain.	A run_pipeline.py that runs section 8.1 in order, stops on the first error and writes a dated log.

When you apply any of these, update this manual in the same change and run section 8.1 with --dry-run first.

11. Coding conventions

Keep the project style when you change or add a script.

```
'''
aida_example.py

   One line that says what the file does.

Purpose

   why the script exists and what it does

Usage

   python aida_example.py --dry-run

Output

   <root>/output/example.csv     one row per paper (CSV)
'''

# ================================================= #
# SECTION NAME
# ================================================= #

# --- subsection
# explanation line 1
# explanation line 2 if applicable
```

- Path variables in capitals (MASTER_BIB_FILE), all other variables in lower case (request_timeout), words joined with underscores.
- Read paths from config.py with a standalone fallback: put SCRIPT_DIR on sys.path, and use FALLBACK_ROOT (AIDA_ROOT, or the folder above scripts) when config is missing.
- Create output folders automatically; offer --dry-run and --verbose; write temporary files and replace, so an open file never leaves half a result.
- Never overwrite a hand edited file or an author's raw file without an explicit option.